

Foraging behaviour of juvenile pink salmon (*Oncorhynchus gorbuscha*) and size-dependent predation risk

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ABSTRACT

Two hypotheses related to effects of juvenile pink salmon (*Oncorhynchus gorbuscha*) foraging behaviour and size on their predation risk were evaluated using field data collected in Prince William Sound, Alaska 1995–97. My results supported the hypothesis that low macrozooplankton density leads to dispersion of juvenile salmon from shallow nearshore habitats and greater predation risk, but zooplankton type was an important factor. When the biomass of large copepods (primarily *Neocalanus* spp.) declined, salmon dispersed from shallow nearshore habitats, and mean daily individual predator consumption of salmon increased by a factor of 5. A concomitant five-fold increase in the probability of occurrence of salmon in predator stomachs supported the notion that increased predation on salmon was caused by a greater overlap between predator and prey when salmon dispersed offshore, not an increase in the number of salmon consumed per feeding bout. The results also generally supported the hypothesis that the timing of predation events modifies the nature of size-dependent predation losses of salmon to different predator groups (small and large planktivores and piscivores). Size-dependent vulnerabilities of salmon to predators were a function of both predator and prey sizes. When simulated predation was shifted from May to June, the vulnerability of salmon became more dependent on their growth than initial size. But, the size- and growth-dependent vulnerabilities of salmon differed more among predator groups than between May and June, suggesting that changes in the composition of predator fields could more strongly affect the nature of size-dependent predation than changes in the timing of predation losses.

Key words: pink salmon, juvenile, foraging behaviour, size-dependent, predation, diet

INTRODUCTION

Examining processes affecting losses of juvenile salmon to various predator taxonomic groups is a necessary first step in understanding mortality. Earlier studies have documented high salmon mortality during the juvenile period without identifying the cause (Parker, 1968; Ricker, 1976; Bax, 1983). Although larval and juvenile fishes are known to be very vulnerable to predators due to their small size (Bailey and Houde, 1989; Sissenwine, 1984; Sogard, 1997), few comprehensive studies have examined the processes affecting predation losses of juvenile salmon (Karpenko, 1998).

In this paper, two hypotheses are evaluated which are related to predation losses of juvenile pink salmon (*Oncorhynchus gorbuscha*) in Prince William Sound (PWS), Alaska. The pink salmon component of the Sound Ecosystem Assessment (SEA) program initially focused on evaluating the 'prey-switching hypothesis' which involved alternative prey modulating predation rates (Willette *et al.*, 2001; this volume p. 14). However, our observations during the course of this 5-year study led to development of two additional hypotheses. The first involved the effect of macrozooplankton density on movement patterns:

H₀: Shallow nearshore habitats provide a refuge from predation for juvenile salmon, but low macrozooplankton density leads to dispersion of salmon from these habitats and greater predation risk.

This conjecture, which was termed the 'refuge-dispersion hypothesis', focused our efforts on juvenile salmon foraging behaviour and the overlap between distributions of predator and prey. Juvenile pink salmon occupy shallow nearshore habitats during the first few weeks of sea life (Cooney *et al.*, 1981; Simenstad *et al.*, 1982). Studies on other species have suggested that these habitats provide a refuge from predation (Helfman, 1993). Walters and Juanes (1993) examined the trade-offs for juvenile fishes that can elect to feed inside or outside of predation refugia. One of the basic premises of their analysis was that juveniles elect to feed outside refugia because food abundance and potential growth rates are higher there. Thus, the motivation to move out of refugia should be related to foraging success, but only if food abundance limits growth.

Similar processes potentially affect juvenile pink salmon in PWS. Large calanoid copepods (primarily

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Neocalanus spp.) are important prey for juvenile pink salmon in PWS (Sturdevant *et al.*, 1993), and these prey are not likely to be abundant nearshore in spring, because they over winter in habitats exceeding 400-m (Fulton, 1973). Competition for limited food resources in nearshore habitats is a concern in PWS, since four hatcheries (Fig. 1) release approximately 500 million juvenile pink salmon into the sound during the spring each year, and about one-half as many wild pink salmon enter the sound from bordering streams.

Two earlier studies have focused on dispersion of juvenile salmon from shallow nearshore habitats and subsequent mortality. Healey (1979) concluded that dispersion of juvenile pink salmon from nearshore habitats was not related to their foraging success based on analysis of stomach fullness data. However, Parsons and LeBrasseur (1973) found that juvenile pink salmon consuming large copepods in the laboratory achieved feeding rates seven times greater than those consuming small copepods, suggesting that the composition of available prey might be important. Healey (1982) concluded that relatively intense size-dependent mortality occurred

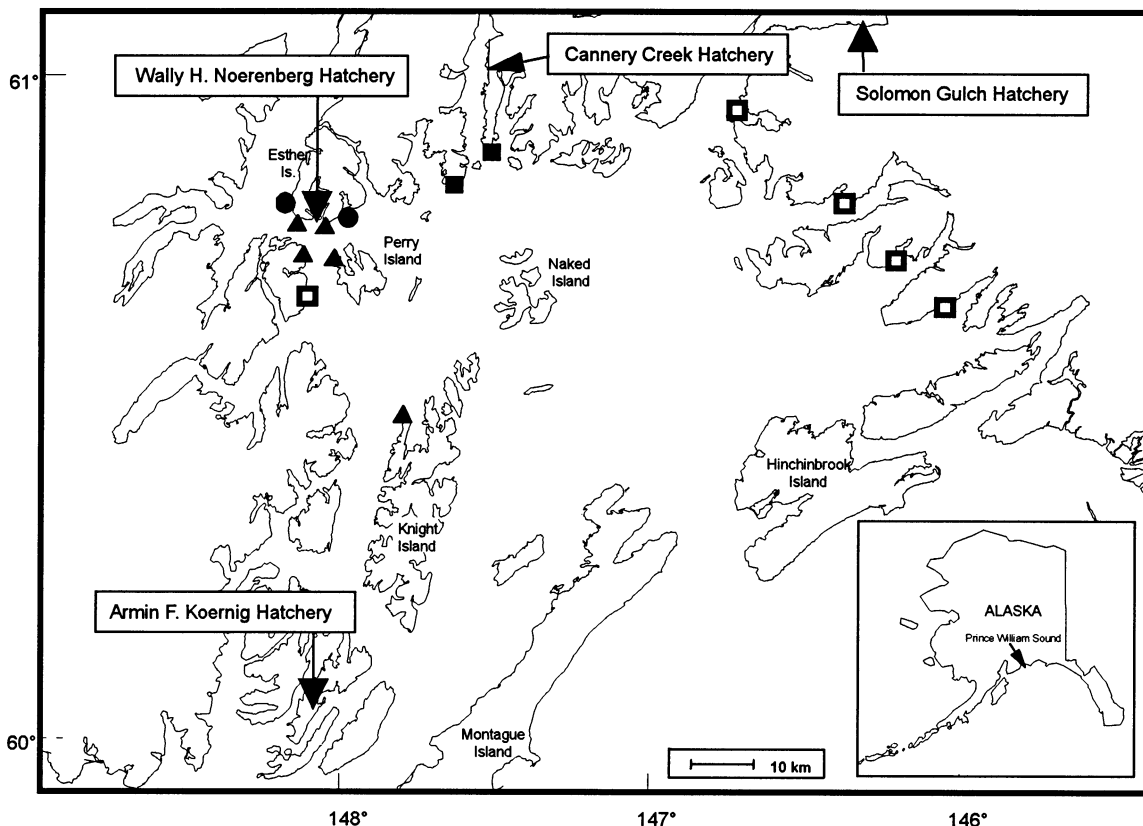
when juvenile chum salmon dispersed offshore. Although Healey (1982) did not identify the cause, he suggested it might be predation.

The second hypothesis involved juvenile salmon size and predation losses:

H₀: Predation on juvenile salmon is size-dependent. The timing of predation modifies the nature of size-dependent losses to different predator groups.

This conjecture was designated the 'size-refuge hypothesis'. Several investigators have found that larger body size upon ocean entry or faster juvenile growth were associated with reduced mortality of juvenile salmon in the field (Holtby *et al.*, 1990; Koenings *et al.*, 1993; Jaenicke *et al.*, 1994; Willette *et al.*, 1999a). Experimental studies examining the mechanisms causing these relationships are limited to juvenile coho salmon preying on pink, chum, and sockeye salmon (Hargreaves and LeBrasseur, 1986; Parker, 1971; Ruggerone, 1989; Ruggerone, 1992). However, similar studies have not been conducted with other predatory fishes, and it is not known whether juvenile coho are important predators on other juvenile salmonids throughout their range. Furthermore, other

Figure 1. Location of the sites sampled in this study and four hatcheries that release juvenile pink salmon in Prince William Sound. Symbol types indicate the years that each site was sampled: solid circles, 1995–97; solid triangles, 1995–96; solid squares, 1996–97; and open squares, 1997.



investigators have found no relationship between juvenile growth and mortality (Fisher and Pearcy, 1988; Matthew and Ishida, 1989; Blackburn, 1990; Willette, 1996).

In the present study, size-dependent vulnerabilities of salmon to different predator groups are examined in relation to initial salmon size upon ocean entry, subsequent growth rate, and the timing of predation. The shape of prey vulnerability functions results from the relative size dependence of encounter rate and capture probability (Bailey and Houde, 1989; Greene, 1986; Juanes, 1994), which are in part a function of relative predator and prey sizes (Miller *et al.*, 1988). Thus, changes in the size composition of predator and prey populations can strongly modify the nature of size-dependent predation losses (Rice *et al.*, 1997). In PWS, a bloom of large copepods typically occurs in May, and a bloom of pteropods sometimes occurs in June (Willette *et al.*, 2001; this volume p. 14). The timing and duration of these blooms (see Eslinger *et al.*, 2001; this volume p. 81) can shift the timing of predation events (Willette *et al.*, 1999a; Willette *et al.*, 2001; this volume), thus potentially modifying size-dependent predation losses of salmon.

My evaluation of the refuge-dispersion hypothesis is divided into three sections. First, data are presented showing spatial and temporal changes in zooplankton biomass and distribution, as well as the distribution and diet composition of juvenile pink salmon in PWS. Whether dispersion of juvenile pink salmon from nearshore habitats is related to their foraging success is evaluated. Second, data are presented showing the spatial and temporal distribution of several potential predators, their food habits, and estimates of their individual daily consumption of salmon. Third, whether juvenile salmon foraging behaviour is related to predation risk is examined.

My evaluation of the size-refuge hypothesis begins with an analysis of the relative size-dependent vulnerabilities of salmon to two functional groups (planktivores and piscivores) and two size classes of predators. The results from these analyses are used to evaluate the relative importance of initial salmon size upon ocean entry and subsequent growth in determining the relative vulnerability of salmon when the timing of predation is shifted from May to June.

METHODS

Field sampling and laboratory analyses

Since the refuge-dispersion hypothesis involved the effects of zooplankton biomass and juvenile salmon relative abundance on salmon foraging behaviour, our sampling program was designed to sample multiple sites exhibiting a range of these values. Knowledge of salmon distribution

obtained from earlier studies in PWS (Willette, 1996) and reconnaissance surveys of the shorelines were used to select sites exhibiting a range of salmon relative abundances. The natural time evolution of the zooplankton bloom provided a range of plankton densities among sites. In 1995–96, nine sites located near hatcheries in western PWS were sampled (Fig. 1). In 1997, six of these same sites were sampled, but four additional sites in eastern and one in western PWS were added to obtain samples from areas where juvenile salmon relative abundances were lower. Most sites were visited from one to three times each year.

A stratified systematic sampling design was used to sample the following parameters at each site: (i) juvenile salmon relative abundance, size composition, and diet composition (ii) predator relative abundance, species and size composition, and diet composition, and (iii) surface-layer macrozooplankton density and composition. Sampling was stratified between nearshore and offshore habitats and by time of day (3-h periods). The 20-m isobath was chosen as the boundary between nearshore and offshore habitats, because kelp beds typically extended to this depth. The 20-m isobath typically occurred within a few hundred metres of the shore at our study sites.

Various gear types were used to sample potential predators on juvenile salmon in different habitats. In offshore areas, a mid-water wing trawl (40 × 28 m) equipped with a net sounder was used to sample potential predators in the 0–60 m layer of the water column. Three gear types were used to sample fishes at two stations within each site. Purse seines (250 × 20 m, 1.5 cm stretch mesh) sampled fishes within 1.5 km of shore and up to the 20-m isobath. Each seine was set in an approximate semi-circle open in the direction of the prevailing current for 20 min. Variable-mesh gill nets (3–12 cm stretch mesh) sampled fishes inshore of the 20-m isobath. Two gill nets were set out from shore perpendicular to the coastline at each station. A small-mesh purse seine (40 × 10 m, 0.5 cm stretch mesh) deployed from a skiff sampled juvenile salmon in shallow nearshore habitats, typically only a few meters deep. At each site, four sets of samples were collected at 3-h intervals between about 21 : 00–09 : 00 h, but in 1995 samples were also collected at 3-h intervals during daylight hours 09 : 00–21 : 00 h. In 1997, no trawl sampling was conducted, and purse seine sampling was not conducted offshore of the 20-m isobath in early May due to lack of availability of a vessel.

Zooplankton samples were collected at two stations located near the 20-m isobath at 3-h intervals to estimate the density of macrozooplankton available to nearshore fishes. In 1995, zooplankton samples were also collected at two stations near the centre of the passages at 3 h intervals to test for differences in zooplankton biomass

between nearshore and offshore habitats. Water depths at these offshore stations exceeded 200 m. Samples were collected with a 0.5 m ring net (243- μ m mesh) towed vertically from a depth of 20 m to the surface. At each site, temperature and salinity were measured with a conductivity–temperature–depth profiler to a depth of 100 m twice each day.

The diet composition of potential predators on salmon was estimated from specimens collected in 1995 ($n = 10\,155$), 1996 ($n = 5289$), and 1997 ($n = 4370$). At least 60 specimens from each fish species were randomly selected from each net set for stomach contents analysis. Fork length and wet weight were measured, and stomachs excised and preserved in formalin for later analysis. In the laboratory, total stomach contents wet weight was measured, and the proportion of total stomach contents in several taxonomic groups was visually estimated (Pearcy *et al.*, 1984). Stomach contents were rinsed into a Petri dish to facilitate identification of prey and estimation of prey composition. This method was used because it allowed a large number of samples to be processed at lower cost. Large sample sizes were necessary because salmon were expected to be a relatively rare prey item in the diets of fish in PWS. Fish found in the stomachs were measured to the nearest millimetre, and all salmon found in the stomachs were weighted to the nearest 0.01 g.

The diet composition of juvenile pink salmon was estimated from specimens collected in 1995 ($n = 3427$),

1996 ($n = 5114$), and 1997 ($n = 2167$). At least 20 specimens of juvenile pink salmon were randomly selected from each net set for stomach contents analysis and preserved in formalin or frozen. In the laboratory, fork length, wet weight, and total stomach content weight were measured. The fractions of the diet composed of large copepods (>2.5 mm), small copepods (<2.5 mm), and 'other' zooplankters were visually estimated (Pearcy *et al.*, 1984), and large copepods in stomachs were enumerated. Zooplankters collected in plankton nets were enumerated into the same three groups using standard subsampling techniques (Coyle *et al.*, 1990). Zooplankton biomass was calculated as the product of abundance and the mean individual wet weight for each group (Coyle *et al.*, 1990).

Distribution and foraging behaviour of juvenile salmon

I evaluated whether dispersion of juvenile pink salmon from shallow nearshore habitats was related to their foraging success by first examining the spatial and temporal distribution of their zooplankton prey, and the diet composition, feeding rate, and distribution of juvenile salmon. Multiple logistic regression analysis was then used to test for relationships between salmon distribution and their diet composition, feeding rate, and prey biomass. Six different types of statistical analyses were conducted (Table 1). Stepwise procedures and multicollinearity tests were applied when appropriate. Data

Table 1. Summary of independent and dependent variables, types of models, sample units and references for each of the statistical tests used to examine the distribution and foraging behaviour of juvenile salmon

Dependent variables	Independent variables	Type of model	Sample unit	References
Mean zooplankton biomass (by zooplankton taxon ^a)	Period Habitat	Two-way ANOVA W/interaction	site $\times$ day $\times$ habitat	Sokal and Rohlf (2000) (p. 328)
Mean zooplankton biomass (by zooplankton taxon)	Period Year	Two-way ANOVA W/interaction	site $\times$ day	Sokal and Rohlf (2000) (p. 328)
Mean salmon diet (by prey taxon)	Period Year Habitat	Three-way factorial ANOVA	site $\times$ day $\times$ habitat	Sokal and Rohlf (2000) (p. 370)
Salmon distribution	Period Year	Fisher exact test	site $\times$ day	Kendall and Stuart (1979) (p. 580)
Salmon distribution	Large copepod biomass Small copepod biomass Other zooplankton biomass Total zooplankton biomass Salmon length	Multiple logistic regression	site $\times$ day	Agresti (1996) (p. 122)
Salmon distribution	Large copepod in diet Small copepod in diet Other zooplankton in diet Stomach fullness Salmon length	Multiple logistic regression	site $\times$ day	Agresti (1996) (p. 122)

^ataxon, taxonomic group.

from all 3 years of the study were available for these analyses.

Analyses of variance (ANOVA) were used to test for differences in biomass of net zooplankton in four taxa between nearshore and offshore habitats and between two periods (early May and early June). The mean biomass of large copepods, small copepods, 'other zooplankton' (mostly pteropods, cladocerans, and various meroplankton), and total zooplankton were independent variables. Samples were only available from 1995 for this analysis. Using samples collected in 1996 and 1997, differences in the biomass of net zooplankton measured near the 20-m isobath among periods (early May, late May, early June) and years were tested for.

Differences in the diet composition and stomach fullness of juvenile pink salmon between shallow nearshore and offshore habitats and among periods and years were also tested for, using the proportion of the diet (arcsin square-root transformed) composed of each taxonomic group of prey and an index of stomach fullness as dependent variables. An index of stomach fullness was estimated using the residuals from a linear regression of mean stomach contents weight (natural logarithm transformed) on juvenile salmon length (Perry *et al.*, 1996).

The distribution of juvenile salmon between shallow nearshore and offshore habitats was examined using a ratio of two indices of salmon relative abundance in these habitats. In shallow nearshore habitats, visual surveys were conducted along the entire shoreline of each study site every 3 h using a skiff. The skiff was driven along a depth contour that allowed the observer to see the bottom, typically only a few metres deep. The observer classified the relative abundance of juvenile salmon as low (=1), moderate (=2), or high (=3). A spotlight was used during night-time surveys. In habitats offshore of the 20-m isobath, an index of salmon relative abundance was derived from catches of juveniles in purse seines. Hartt (1980) described the method we used to set the nets and estimate relative abundance. The geometric mean catch per net set (GMC) of juvenile salmon was calculated for each site, and a categorical index of salmon relative abundance (low = 1, moderate = 2, high = 3) was created by partitioning the GMC data from all sites at the 33rd and 66th percentiles. The ratio of these two indices of salmon relative abundance was used to establish a class variable (R_a) indicating the relative distribution of salmon between shallow nearshore and offshore habitats at each site, i.e. if the relative abundance index offshore was less than nearshore ($R_a = 1$), if the indices were equal ($R_a = 2$), and if the offshore index was greater than nearshore ($R_a = 3$). Three separate Fisher exact tests were conducted to test for a difference in the frequency of occurrence of R_a between periods within each year.

Multiple logistic regression analyses were conducted to examine the relationship between salmon distribution and salmon foraging success using data from all 3 years of the study. In separate analyses, the frequency of occurrence of R_a was the dependent variable, and net zooplankton (measured near 20-m isobath) and salmon diet (measured in habitats a few meters deep) data were independent variables indexing salmon prey biomass and foraging success nearshore. In the first analysis, five independent class variables were derived by partitioning mean salmon length, large copepod biomass, small copepod biomass, 'other zooplankton' biomass, and total zooplankton biomass data into five classes approximately dividing the total number of observations equally. In the second analysis, the same procedure was used to establish five independent class variables derived from mean salmon length, an index of salmon stomach fullness, and the proportion of salmon diets composed of large copepods, small copepods, and 'other zooplankton'. These procedures were used to provide for calculation of frequencies of R_a at various levels of the independent class variables (Sokal and Rohlf, 2000). Salmon length was included as an independent variable because LeBrasseur and Parker (1964) observed an ontogenetic migration of juvenile salmon from nearshore to offshore habitats. Three prey types were included as independent variables because laboratory studies have shown that salmon feeding rates differ among prey types (Parsons and LeBrasseur, 1973). A backward selection procedure was used to identify those independent variables that were most highly correlated with the dependent variable.

Distribution and food habits of potential salmon predators

Three types of statistical analyses were conducted to examine spatial and temporal changes in the distribution of potential predators on salmon, their food habits and daily consumption of salmon (Table 2). Stepwise procedures and multicollinearity tests were applied when appropriate. Data from all 3 years of the study were used in these analyses.

Potential predators on salmon were placed into nine taxonomic groups: herring, immature (<30 cm) and adult (>30 cm) walleye pollock, other gadids (*Gadus macrocephalus*, *Microgadus proximus*), adult salmon (*Oncorhynchus* spp.), trout (*Salvelinus malma*, *O. mykiss*), benthic fishes (Cottidae, Hexagrammidae, Stichaeidae, Cyclopteridae, Zoarcidae, Bathymasteridae), pelagic rockfishes (*Sebastes melanops*, *S. ciliatus*, *S. flavidus*) and demersal rockfishes (*Sebastes nebulosus*, *S. caurinus*, *S. maliger*, *S. rubberimus*, and others). Several other taxonomic groups of fishes (Osmeridae, Ammodytidae, Gasterosteidae, Zaproridae, Pleuronectidae, Anarhichadidae, Anoplopomatidae, and Lamnidae) were not included in the

Table 2. Summary of independent and dependent variables, types of models, sample units and references for each of the statistical tests used to examine the distribution and food habits of potential salmon predators

Dependent variables	Independent variables	Type of model	Sample unit	References
Mean predator catch (by predator taxon ^a)	Period Year	Two-way ANOVA W/interaction	site × day × predatory taxon	Sokal and Rohlf (2000) (p. 328)
Mean diet proportion (by predator and prey taxon)	Period Year	Two-way ANOVA W/interaction	site × day × predatory taxon	Sokal and Rohlf (2000) (p. 328)
Mean daily consumption of salmon (by predator taxon)	Period Year	Two-way ANOVA W/interaction	site × day × predatory taxon	Sokal and Rohlf (2000) (p. 328)

^ataxon, taxonomic group.

analysis, because they were relatively rare in catches and did not consume significant numbers of juvenile salmon.

Analyses of net catch data were conducted to examine spatial and temporal changes in relative abundance of the nine taxonomic groups of potential predators on salmon (Table 2). Data from mid-water trawls were used for adult pollock, purse seines for adult salmon and herring, and variable-mesh gillnets for the other six taxonomic groups. The gear type used to represent the relative abundance of each group consistently caught the greatest numbers of that group. The dependent variable in each model was the mean catch per set (natural logarithm transformed), except that catch per hour of tow was used for adult pollock captured in trawls.

Analyses of stomach content data were conducted to examine spatial and temporal changes in the diet composition of potential predators on salmon (Table 2). Five separate ANOVAs were conducted for each of six taxonomic groups of potential salmon predators. Herring and pollock were excluded from the analysis because Willette *et al.* (1999a) had already described their diets in PWS. For each predator group, the dependent variables in the five separate models were the mean proportion of the diet composed of large calanoid copepods, other macrozooplankton, benthic invertebrates, juvenile salmon and other fishes, weighted by the catch of each predator in each net set. A weighted mean was used because samples taken from large catches represented a larger number of animals and should therefore be weighted accordingly. The weighted mean diet proportion ($\bar{P}_w$) for prey type i in a specific predator group sampled at a site on a specific day (sample unit) was estimated from

$$\bar{P}_w = \sum_{l=1}^m \left(\frac{\sum_{k=1}^n \left(\frac{w_{ik} \cdot c_k}{w_{Tk} \cdot c_k} \right)}{m} \right) \quad (1)$$

where $w_{ik} = \sum_{j=1}^n w_{ij}$ was the wet weight of prey type i summed among j individual predators in net set k within time of day (3-h) period l , $w_{Tk} = \sum_{j=1}^n w_{Tj}$ was the wet weight of all prey summed among j individual predators in net set k , and c_k was the catch of the specific predator group in net set k . The wet weight of prey type i in individual predator stomachs (w_{ij}) was estimated from $w_{ij} = p_{ij} \cdot w_{Tj}$, where p_{ij} was the visually estimated proportion of total stomach contents composed of prey type i in an individual predator stomach.

Nine separate ANOVAs were conducted to examine spatial and temporal changes in the daily individual consumption of juvenile salmon by each of nine taxonomic groups of potential salmon predators (Table 2). Mean daily individual predator consumption of juvenile salmon (number day⁻¹ predator⁻¹) was estimated from $I = (24\bar{S}_w\alpha)/\bar{w}$, where $\bar{S}_w$ was the mean wet weight of juvenile salmon in the predator stomachs weighted by the catch of each predator group in each net set, α was the gastric evacuation rate, and $\bar{w}$ was the mean wet weight of whole individual juvenile salmon found in the stomachs (Eggers, 1977). The weighted mean wet weight of juvenile salmon in the stomachs of a specific predator group sampled at a site on a specific day (sample unit) was estimated from

$$\bar{S}_w = \sum_{l=1}^m \left(\frac{\sum_{k=1}^n \bar{w}_k \cdot c_k}{m} \right) / c_T \quad (2)$$

where $\bar{w}_k$ was the mean wet weight of salmon in j individual predators in net set k within time of day (3-h) period l , and c_T was the total catch of a specific predator group summed among k net sets within time of day (3-h) period l . Temperature and prey size-dependent gastric evacuation rates were estimated using data provided by He and Wurtsbaugh (1993). Gastric evacuation rates were estimated using mean 0–60 m layer temperatures for adult

pollock captured in trawls offshore, and mean 0–20 m layer temperatures for the other predators captured in seines or gillnets. Stomach content data for the period from 21 : 00 to 09 : 00 h the following morning were used in these calculations because data were available from this time period from all sites. An ANOVA conducted using data from 1995 indicated that the mean number of salmon in the stomachs of seven of the predator groups was not significantly different between this 12-h period and the full 24-h day. Demersal and pelagic rockfishes were not included in the analysis due to lack of sufficient data.

Juvenile salmon foraging behaviour and predation risk

The hypothesis that dispersion of juvenile salmon from shallow nearshore habitats led to greater predation risk was evaluated by examining relationships between salmon distribution and (i) the frequency of occurrence of salmon in predator stomachs; (ii) the mean number of salmon consumed by individual predators during feeding bouts (contagious feeding); and (iii) the mean daily number of salmon consumed by individual predators. If dispersion from shallow nearshore habitats resulted in a greater overlap between predator and prey distributions, the frequency of occurrence of salmon in predator stomachs should increase (assuming no change in predator abundance). Conversely, dispersion may result in predators attacking schools of vulnerable salmon, causing an increase in the number of salmon consumed per predator feeding bout. Three different statistical analyses were conducted (Table 3). Stepwise procedures and multicollinearity tests were applied when appropriate. Data from all 3 years of the study were used in these analyses.

A multiple logistic regression analysis was conducted to examine the relationship between the frequency of occurrence of salmon in predator stomachs and salmon distribution. The independent variables were salmon

distribution (R_a), mean juvenile salmon length measured in shallow nearshore habitats, and predator functional group. A class variable was derived for salmon length by partitioning mean length data into five classes approximately dividing the total number of observations equally. Three predator functional groups were established based upon their diets: facultative planktivores (pollock, herring, and pelagic rockfishes), piscivores (gadids and trout), and benthic invertebrate feeders (benthic fishes and demersal rockfishes). For brevity, facultative planktivores will be called 'planktivores' and benthic invertebrate feeders will be called 'invertebrate feeders'.

An ANCOVA was conducted to examine the relationship between the mean number of salmon consumed per feeding bout and salmon distribution. Only predators with juvenile salmon in their stomachs that were estimated to have captured the salmon within 2 h of collection of the predator were included in the analysis. The length of salmon in predator stomachs was measured, and a length–weight regression derived from net-caught salmon was used to estimate the wet weight of the salmon prey at the time they were captured by the predator. The mean time since prey capture (t) was then estimated from mean prey weight at capture ($(\bar{w}_t)$), mean initial prey weight ($(\bar{w}_0)$, back calculated from length), and temperature-specific gastric evacuation rate (α), i.e. $t = \frac{\log(\bar{w}_t) - \log(\bar{w}_0)}{\alpha}$. The mean number of juvenile salmon consumed per feeding bout (natural logarithm transformed) was the dependent variable in the model. The same independent variables were used as in the previous analysis (Table 3).

Finally, an ANCOVA was conducted to examine the relationship between the mean daily number of salmon consumed by individual predators (I) and salmon distribution. The dependent variable was natural logarithm

Table 3. Summary of independent and dependent variables, types of models, sample units and references for each of the statistical tests used to examine juvenile salmon foraging behaviour and predation risk

Dependent variables	Independent variables	Type of model	Sample unit	References
Frequency of occurrence of salmon in predator stomachs	Salmon dist ^a Salmon length Predator group	Multiple logistic regression	site × day × predatory taxon ^b	Agresti (1996) (p. 122)
Mean number of salmon per feeding bout	Salmon dist Salmon length Predator group	Two-way ANCOVA W/interaction	site × day × predatory taxon	Sokal and Rohlf (2000) (p. 503)
Mean daily predator consumption of salmon predator group	Salmon dist Salmon length	Two-way ANCOVA W/interaction	site × day × predatory taxon	Sokal and Rohlf (2000) (p. 503)

^adist, distribution.

^btaxon, taxonomic group.

transformed, and the same independent variables were used as in the previous analysis (Table 3).

Size-dependent vulnerability of salmon and timing of predation

Four analyses were conducted to examine whether the relative vulnerabilities of salmon to two predator functional groups (planktivores and piscivores) were size dependent. A generalized reduced gradient nonlinear optimization algorithm (Frontline Systems, Inc., Incline Village, NV 89450) was used to calculate relative length-specific vulnerabilities (Pv_i) of salmon to small planktivores (<30 cm), large planktivores (>30 cm), small piscivores and large piscivores. Invertebrate feeders were excluded from the analysis due to insufficient data. Pv_i was calculated over i salmon-length intervals and three 2-week periods (early May, late May, and early June). The algorithm calculated the values of Pv_i that minimized the sums of squares (SSQ) given by $SSQ = \sum \hat{P}s_{ij} - P s_{ij}$, where $\hat{P}s_{ij}$ was the expected distribution of juvenile salmon lengths in predator stomachs during date period j , and $P s_{ij}$ was the actual distribution of salmon lengths observed in predator stomachs during date period j . The expected distribution of salmon lengths in stomachs was given by $\hat{P}s_{ij} = \frac{Pd_{ij}}{\sum Pd_{ij}}$, where Pd_{ij} was the probability of a salmon of length i being included in the diet during date period j . Pd_{ij} was given by $Pd_{ij} = Pv_i \cdot Pe_{ij}$, where Pe_{ij} was the distribution of juvenile salmon fork lengths in the environment during date period j estimated from seine samples. Frequencies were calculated by aggregating juvenile salmon length data ranging from 2.5 to 7.5 cm into five 1-cm intervals. A bootstrap technique was used to calculate 95% confidence intervals on Pv_i and evaluate whether parameter estimates differed from one another. Data from all 3 years of the study were used in these analyses.

Several analyses were conducted to examine the relative importance of initial salmon length upon ocean entry and subsequent growth in determining the vulnerabilities of salmon to small and large planktivores and piscivores when the timing of predation was shifted from May to June. Non-linear regression analysis was first used to model the vulnerability of salmon to each predator group in relation to salmon length. The dependent variables in the regression models were the discrete prey vulnerabilities estimated from the previous analyses. These size-dependent prey vulnerability functions were then applied to simulated salmon growth curves (in terms of length) over the initial 60 days of sea life (assumed to be 1 May–30 June) to derive a mean vulnerability over the entire period. The initial salmon lengths (2–6 cm) and growth rates (0.5–2.5% length day⁻¹) used in the simula-

tions were within the ranges observed in PWS. For each predator group, two sets of simulations were conducted with predation occurring in May and June, respectively.

RESULTS

Distribution and foraging behaviour of juvenile salmon

The biomass of large copepods declined significantly from early May to June 1995 in both nearshore and offshore habitats (Table 4). The least-squares mean biomass of large copepods was also significantly greater ($P = 0.040$) in offshore (mean = 0.19) than nearshore (mean = 0.10) habitats in both months. The biomass of 'other zooplankton' increased significantly in offshore habitats from early May to June, 1995 (Table 4), but there were no other significant differences in zooplankton biomass between periods or habitats and the interaction terms in the models were not significant. In 1996, the nearshore biomass of large copepods again declined significantly from early May to June, while the biomass of 'other zooplankton' increased significantly over this same period. In 1997, the nearshore biomass of large copepods was low during all date periods, but the biomass of 'other zooplankton' again increased from late May to early June. The biomass of 'other zooplankton' was also significantly ($P < 0.001$) different among years, and the pattern of change in biomass of large copepods among date periods differed significantly ($P = 0.003$) among years.

In 1995, changes in juvenile pink salmon diet composition among periods and habitats were similar to those observed in the zooplankton. Stomach fullness and the proportion of the diet composed of large copepods declined significantly in nearshore habitats from early May to June, while the proportion of the diet composed of 'other zooplankton' increased significantly in both habitats during this same period (Table 5). Similar changes in diet composition and stomach fullness among periods and habitats were apparent in 1996, but not in 1997. The interaction terms were not significant in any of the models, but changes in diet composition and stomach fullness were at least marginally significant among habitats, periods, and years in all cases. Between habitats, stomach fullness was greater ($P = 0.055$) offshore (mean = 0.002) than nearshore (mean = -0.001), large copepods composed a greater ($P = 0.087$) fraction of the diet offshore (mean = 26%) than nearshore (mean = 19%), small copepods composed a lesser ($P < 0.001$) fraction of the diet offshore (mean = 17%) than nearshore (mean = 34%), and 'other zooplankton' composed a greater ($P = 0.005$) fraction of the diet offshore (mean = 57%) than nearshore (mean = 45%). Excluding 1997, when the bloom of large copepods in early May did not

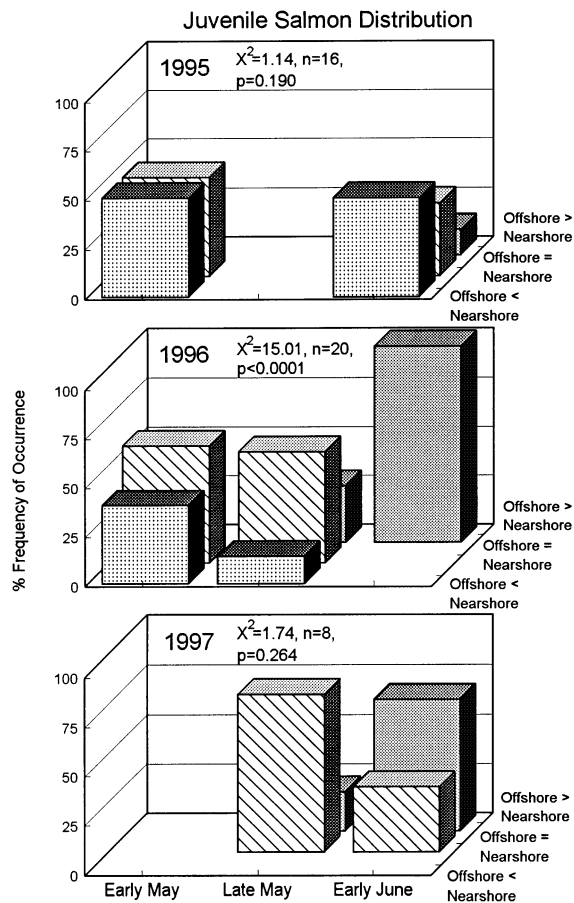
Table 4. Least-squares mean biomass (g m^{-3}) of large copepods, small copepods, and 'other' zooplankton in nearshore and offshore habitats of Prince William Sound during early May, late May, and early June, 1995–97. The standard deviation of the mean is indicated in parentheses. Statistical tests are for a difference in mean diet composition among date periods. Samples collected to characterize zooplankton biomass in nearshore habitats were collected near the 20-m isobath. Statistically significant *P*-values are indicated in bold

Year	<i>n</i>	Variable	Habitat	Early May	Late May	Early June	<i>P</i> -value
1995	16	Large copepods	Nearshore	0.17 (0.05)	–	0.03 (0.04)	0.015
			Offshore	0.27 (0.04)	–	0.12 (0.04)	0.033
		Small copepods	Nearshore	0.23 (0.07)	–	0.20 (0.06)	0.599
			Offshore	0.19 (0.05)	–	0.31 (0.06)	0.192
		Other zooplankton	Nearshore	0.19 (0.05)	–	0.25 (0.04)	0.386
			Offshore	0.16 (0.04)	–	0.33 (0.04)	0.009
		Total zooplankton	Nearshore	0.59 (0.13)	–	0.48 (0.12)	0.212
			Offshore	0.62 (0.10)	–	0.77 (0.11)	0.276
1996	20	Large copepods	Nearshore	0.18 (0.04)	0.04 (0.03)	0.01 (0.03)	< 0.001
		Small copepods	Nearshore	0.26 (0.07)	0.26 (0.06)	0.31 (0.06)	0.819
		Other zooplankton	Nearshore	0.02 (0.03)	0.06 (0.03)	0.17 (0.02)	< 0.001
		Total zooplankton	Nearshore	0.46 (0.11)	0.36 (0.09)	0.49 (0.09)	0.276
1997	24	Large copepods	Nearshore	0.02 (0.03)	0.06 (0.02)	0.06 (0.02)	0.148
		Small copepods	Nearshore	0.18 (0.08)	0.24 (0.04)	0.32 (0.06)	0.316
		Other zooplankton	Nearshore	0.07 (0.03)	0.07 (0.02)	0.15 (0.03)	0.021
		Total zooplankton	Nearshore	0.27 (0.11)	0.37 (0.07)	0.53 (0.08)	0.080

Table 5. Least-squares mean stomach fullness index and diet composition (%) for juvenile pink salmon in nearshore and offshore habitats of Prince William Sound during early May, late May, and early June, 1995–97. The standard deviation of the mean is indicated in parentheses. Statistical tests are for a difference in mean diet composition among date periods. Statistically significant *P*-values are indicated in bold

Year	<i>n</i>	Variable	Habitat	Early May	Late May	Early June	<i>P</i> -value
1995	16	Large copepods	Nearshore	46.4 (5.0)	–	12.5 (5.0)	0.002
			Offshore	64.3 (8.2)	–	20.0 (5.8)	0.096
		Small copepods	Nearshore	38.1 (5.3)	–	43.3 (5.3)	0.594
			Offshore	18.4 (8.7)	–	10.7 (6.2)	0.242
		Other zooplankton	Nearshore	13.7 (5.7)	–	43.8 (5.7)	< 0.001
			Offshore	17.0 (9.3)	–	69.0 (6.6)	0.054
		Stomach fullness index	Nearshore	0.005 (0.003)	–	–0.006 (0.003)	0.035
			Offshore	0.007 (0.005)	–	0.007 (0.004)	0.980
1996	20	Large copepods	Nearshore	50.7 (6.4)	8.4 (5.4)	1.2 (5.0)	< 0.001
			Offshore	65.0 (8.2)	14.8 (6.4)	4.8 (5.4)	< 0.001
		Small copepods	Nearshore	25.9 (6.7)	47.1 (5.7)	32.4 (5.3)	0.030
			Offshore	12.8 (8.7)	19.0 (6.7)	17.0 (5.7)	0.877
		Other zooplankton	Nearshore	22.7 (7.2)	42.9 (6.1)	65.9 (5.7)	< 0.001
			Offshore	22.1 (7.2)	65.8 (7.2)	78.1 (6.1)	< 0.001
		Stomach fullness index	Nearshore	0.004 (0.004)	–0.005 (0.003)	–0.013 (0.003)	< 0.001
			Offshore	0.001 (0.005)	–0.008 (0.004)	–0.007 (0.003)	0.362
1997	18	Large copepods	Nearshore	0.0 (3.5)	2.0 (4.7)	1.8 (7.1)	0.706
			Offshore	–	0.1 (8.2)	0.2 (5.4)	0.858
		Small copepods	Nearshore	22.5 (17.5)	30.1 (5.0)	15.1 (7.5)	0.236
			Offshore	–	29.4 (8.7)	15.3 (5.7)	0.180
		Other zooplankton	Nearshore	77.5 (20.3)	67.3 (5.4)	83.0 (8.1)	0.282
			Offshore	–	70.2 (9.3)	84.4 (6.1)	0.183
		Stomach fullness index	Nearshore	–0.006 (0.006)	–0.003 (0.003)	0.000 (0.005)	0.364
			Offshore	–	0.000 (0.005)	0.002 (0.003)	0.833

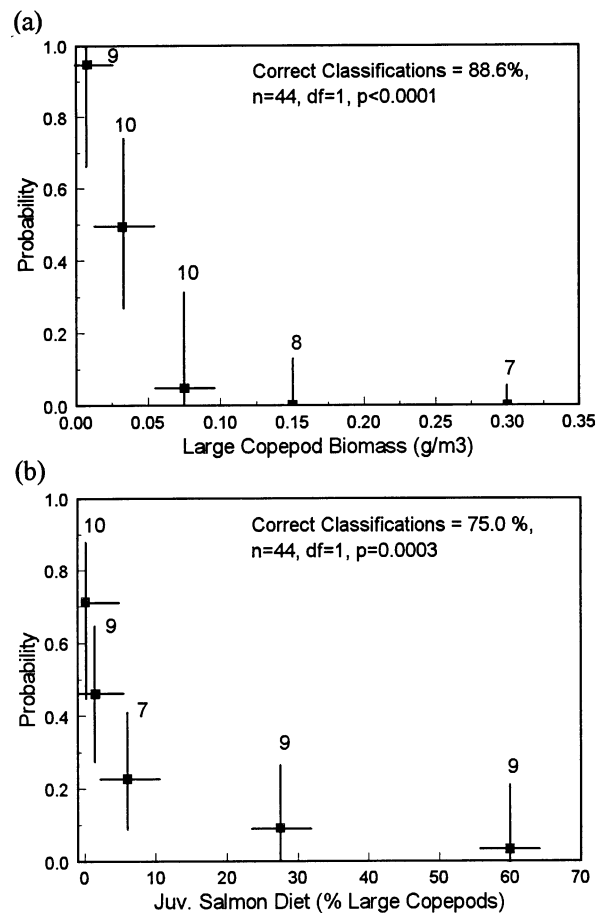
Figure 2. Percent frequency of occurrence of a ratio of juvenile salmon relative abundance in nearshore and offshore habitats (R_a) in Prince William Sound during early May, late May, and early June, 1995–97.



occur (Table 4), the fraction of the diet composed of large copepods was significantly greater ($P = 0.047$) offshore (mean = 31%) than nearshore (mean = 22%). The 'other zooplankton' prey group was composed of 50% meroplankton (including fish larvae), 44% pteropods, cladocerans, and ostracods, and 6% amphipods and euphausiids. Thus, juveniles feeding offshore in June largely consumed meroplankton, whereas those that remained nearshore consumed more small copepods.

An examination of the spatial and temporal changes in juvenile salmon distribution indicated that salmon tended to shift from a primarily nearshore to an offshore distribution between early May and June, but the frequency of occurrence of R_a was only significantly different among periods in 1996 (Fig. 2). Although salmon species could not be determined from visual shoreline surveys, net catch data indicated the species composition of juvenile salmon aggregated across all study sites was 95% pink salmon and 5% chum salmon (*O. keta*).

Figure 3. (a) Relationship between the biomass of large copepods measured at the 20-m isobath and the probability that juvenile salmon relative abundances were greater in offshore than nearshore habitats as predicted from logistic regression analysis. Vertical bars represent the 95% confidence intervals on the estimated probabilities, and numbers above the vertical bars indicate sample sizes. (b) Relationship between the percent of juvenile salmon diets composed of large copepods in nearshore habitats (inshore of 20-m isobath) and the probability that juvenile salmon relative abundances were greater in offshore than nearshore habitats as predicted from logistic regression analysis. Vertical bars represent the 95% confidence intervals on the estimated probabilities, and numbers above the vertical bars indicate sample sizes.



The notion that dispersion of juvenile pink salmon from shallow nearshore habitats was related to their foraging success in these habitats was supported by results obtained from two separate multiple logistic regression analyses. The results indicated that prey type was an important factor affecting foraging success. In the first analysis, changes in the distribution of salmon between shallow nearshore and offshore habitats were related to large copepod biomass measured at the 20-m isobath (Fig. 3a).

Table 6. Least-squares mean catch per net set for four taxonomic groups of fishes in Prince William Sound during early May, late May, and early June, 1995–97. The standard deviation of the mean is indicated in parentheses. Statistical tests are for a difference in mean catch per net set among date periods. Each gear sampled a different habitat: trawls-offshore of 20-m isobath 0–60 m layer, seines-offshore of 20-m isobath 0–20 m layer, and gillnets-inshore of 20-m isobath. Statistically significant *P*-values are indicated in bold

Predator taxon	<i>n</i>	Gear	Early May	Late May	Early June	<i>P</i> -value
Adult pollock (> 30 cm)	38	Trawls	67.22 (0.50)	280.48 (0.92)	36.54 (0.48)	0.040
Immature pollock (<30 cm)	60	Gillnets	0.47 (0.13)	0.76 (0.17)	1.26 (0.12)	0.039
Salmon	51	Seines	12.57 (0.71)	21.48 (0.84)	117.19 (0.51)	0.004
Trout	60	Gillnets	0.02 (0.07)	0.07 (0.09)	0.72 (0.07)	<0.001

Table 7. Least-squares mean catch per net set for seven taxonomic groups of fishes in Prince William Sound during 1995–97. The standard deviation of the mean is indicated in parentheses. Statistical tests are for a difference in mean catch per net set among years. Each gear sampled a different habitat: trawls-offshore of 20-m isobath 0–60 m layer, seines-offshore of 20-m isobath 0–20 m layer, and gillnets-inshore of 20-m isobath. Statistically significant *P*-values are indicated in bold

Predator taxon	<i>n</i>	Gear	1995	1996	1997	<i>P</i> -value
Herring	51	Seines	45.61 (0.76)	6.29 (0.58)	149.76 (0.82)	<0.001
Adult pollock (> 30 cm)	38	Trawls	181.14 (0.56)	43.15 (0.41)	–	0.017
Gadids	60	Gillnets	0.48 (0.12)	1.32 (0.09)	0.73 (0.08)	0.006
Trout	60	Gillnets	0.03 (0.09)	0.38 (0.07)	0.34 (0.06)	0.029
Benthic fishes	60	Gillnets	1.59 (1.58)	1.92 (1.91)	1.33 (1.34)	0.017
Pelagic rockfishes	60	Gillnets	0.26 (0.16)	0.92 (0.12)	0.71 (0.11)	0.095
Demersal rockfishes	60	Gillnets	0.65 (0.12)	1.50 (0.16)	1.79 (0.15)	0.080

In the second analysis, changes in the distribution of salmon between these habitats were related to the proportion of salmon diets composed of large copepods measured in shallow nearshore habitats (Fig. 3b). Changes in salmon distribution were not related to salmon length, or the biomass or consumption of other prey types, or stomach fullness.

Distribution and food habits of potential salmon predators

The relative abundance of immature pollock and trout in nearshore habitats increased significantly from early May to June (Table 6). In offshore habitats, the relative abundance of adult salmon increased while abundances of adult pollock decreased during this same period (Table 6). Least-squares mean catch per net set increased by about an order of magnitude for trout and salmon and by a factor of 2 for immature pollock. Least-squares mean catch of pollock per hour of tow declined by about a factor of 2 from early May to June. The relative abundance of five taxonomic groups of predators also differed significantly among years (Table 7), but the interaction terms in the models were not significant for any of the predator groups, indicating that changes in relative abundance among periods did not differ among years.

Juvenile salmon and other fishes dominated the diets of trout and gadids, but there were no seasonal changes in the amounts of fish consumed by these two predators (Table 8). Benthic invertebrates comprised the majority of the diets of demersal rockfishes and benthic fishes. Pelagic rockfishes primarily consumed macrozooplankton and large copepods, while adult salmon largely consumed macrozooplankton. The period–year interaction terms were significant in several of these models, so the least-squares means presented in Table 8 indicate the marginal means that would be expected had the design been balanced. Mean daily individual predator consumption of salmon was significantly different among years for adult pollock ($P = 0.011$) and pelagic rockfishes ($P = 0.040$), but none of the other terms in these models were significant. The mean daily individual predator consumption of salmon was significantly greater for trout and gadids ($P < 0.003$) compared with the other taxonomic groups of predators (Table 9).

Juvenile salmon foraging behaviour and predation risk

Dispersion of salmon from shallow nearshore habitats was associated with an increase in the frequency of occurrence of salmon in predator stomachs but not an increase in the number of salmon consumed per feeding

Table 8. Least-squares mean diet composition (%) for six taxonomic groups of fishes in Prince William Sound during early May, late May, and early June, 1995–97. The standard deviation of the mean is indicated in parentheses. Statistical tests are for a difference in mean diet composition among date periods. Statistically significant *P*-values are indicated in bold

Predator taxonomic group	n	Prey taxonomic group	Early May	Late May	Early June	<i>P</i> -value
Gadids	58	Large copepod	2.9 (0.9)	0.1 (0.9)	0.3 (0.7)	0.091
		Macrozooplankton	26.2 (5.2)	26.0 (4.7)	35.3 (3.8)	0.042
		Benthic invertebrate	22.8 (4.9)	22.1 (4.5)	20.6 (3.5)	0.656
		Juvenile salmon	16.8 (4.9)	11.7 (6.0)	7.8 (4.4)	0.371
		Other fish	30.0 (6.4)	31.0 (7.8)	32.6 (5.6)	0.952
Salmon	49	Large copepod	0.4 (0.3)	0.1 (0.3)	0.1 (0.1)	0.888
		Macrozooplankton	49.5 (12.0)	60.4 (11.8)	88.4 (5.8)	0.013
		Benthic invertebrate	12.5 (5.2)	3.3 (5.2)	3.0 (2.6)	0.552
		Juvenile salmon	15.4 (7.3)	3.2 (7.3)	0.0 (3.6)	0.181
		Other fish	20.4 (10.4)	33.0 (10.3)	2.9 (5.1)	0.042
Trout	38	Large copepod	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.881
		Macrozooplankton	6.7 (7.5)	10.3 (5.0)	13.1 (3.0)	0.638
		Benthic invertebrate	38.9 (12.9)	14.6 (8.6)	5.1 (5.2)	0.053
		Juvenile salmon	1.7 (16.3)	37.2 (10.9)	27.9 (6.5)	0.179
		Other fish	53.5 (16.6)	37.8 (11.1)	51.3 (6.6)	0.621
Benthic fishes	60	Large copepod	0.1 (0.0)	0.0 (0.0)	0.0 (0.0)	0.301
		Macrozooplankton	16.5 (1.9)	10.7 (2.5)	11.6 (1.8)	0.085
		Benthic invertebrate	51.6 (3.9)	58.3 (5.0)	45.8 (3.7)	0.193
		Juvenile salmon	0.2 (1.0)	0.0 (1.3)	2.5 (0.9)	0.074
		Other fish	24.5 (3.7)	26.2 (4.8)	32.0 (3.5)	0.243
Pelagic rockfishes	45	Large copepod	39.8 (5.3)	13.3 (6.0)	0.7 (4.3)	< 0.001
		Macrozooplankton	54.6 (8.0)	58.4 (9.1)	57.7 (6.4)	0.941
		Benthic invertebrate	3.6 (1.8)	2.7 (2.0)	3.8 (1.4)	0.842
		Juvenile salmon	0.0 (1.5)	1.9 (1.7)	3.1 (1.2)	0.229
		Other fish	0.6 (6.0)	16.4 (6.8)	23.0 (4.8)	0.008
Demersal rockfishes	58	Large copepod	3.8 (1.6)	3.3 (2.0)	1.5 (1.6)	0.546
		Macrozooplankton	29.8 (5.6)	25.2 (7.0)	21.3 (5.3)	0.700
		Benthic invertebrate	41.4 (5.7)	49.9 (7.2)	50.5 (5.5)	0.498
		Juvenile salmon	0.4 (0.9)	0.0 (1.2)	1.8 (0.9)	0.275
		Other fish	15.2 (3.7)	18.1 (4.7)	18.6 (3.6)	0.610

bout. The frequency of occurrence of salmon in predator stomachs was significantly correlated ($P < 0.0001$) with salmon distribution, mean salmon length, and predator functional group (Fig. 4). The probability that salmon were present in predator stomachs was inversely correlated with salmon length for all three predator groups, but the slope of the relationship was less for planktivores than the other two predators. The probability that salmon were present in stomachs was also much greater for piscivores than the other two predators. As juvenile salmon dispersed from shallow habitats, the probability of salmon being present in predator stomachs increased by about a factor of 5 for all three predator groups (Fig. 4).

The mean number of salmon consumed per feeding bout was not correlated with salmon distribution but was significantly correlated ($P < 0.001$) with mean salmon

length and predator functional group (Fig. 5a). The least-squares mean number of salmon consumed per bout was significantly greater ($P < 0.015$) for piscivores (mean = 4.73) than planktivores (mean = 2.32) or invertebrate feeders (mean = 2.05). The interaction term between the two independent variables was also significant, indicating that the slope of the relationship between mean salmon length and number of salmon consumed per feeding bout was only different from zero (negative slope) for piscivores.

Dispersion of salmon from shallow nearshore habitats was associated with a five-fold increase in the mean daily individual predator consumption of salmon. The mean daily consumption of salmon was significantly correlated with salmon distribution, mean salmon length, and predator functional group (Fig. 5b). The least-squares mean

Table 9. Mean daily individual predator consumption of juvenile salmon (number day⁻¹) for nine taxonomic groups of fishes in Prince William Sound during May and June, 1995–97. The standard error of the mean is indicated in parentheses. Mean consumption is not significantly different among taxonomic groups with the same code

Predator taxon	n	Mean daily consumption	Code
Herring	61	0.017 (0.085)	1
Adult pollock (>30 cm)	37	0.132 (0.122)	1
Immature pollock (<30 cm)	56	0.119 (0.091)	1
Gadids	58	0.989 (0.087)	2
Salmon	49	0.126 (0.114)	1
Trout	38	1.991 (0.109)	2
Benthic fishes	60	0.113 (0.085)	1
Pelagic rockfishes	45	0.133 (0.098)	1
Demersal rockfishes	58	0.027 (0.088)	1

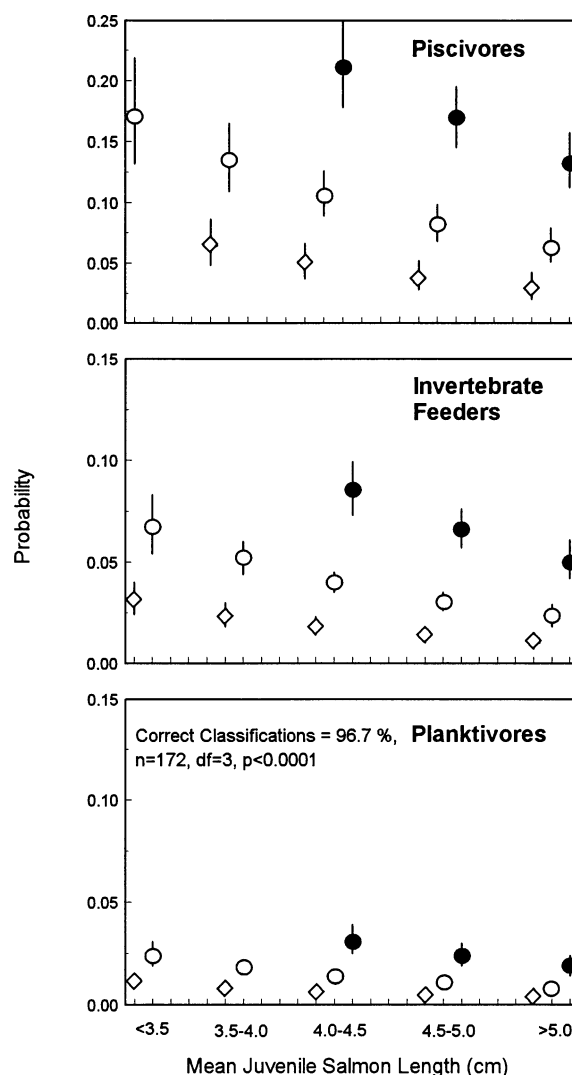
daily consumption of salmon was significantly greater ($P = 0.003$) when salmon were more abundant offshore than nearshore ($R_a = 3$: mean = 1.32; $R_a = 2$: mean = 0.22; $R_a = 1$: mean = 0.24). The least-squares mean daily consumption of salmon was also significantly greater ($P < 0.001$) for piscivores (mean = 1.55) than planktivores (mean = 0.21) or invertebrate feeders (mean = 0.14). Two interaction terms were significant ($P < 0.030$) in the model. The first interaction term indicated that the daily number of salmon consumed was more size dependent (negative slope) for piscivores than the other two predator groups. The second interaction term indicated that the number of salmon consumed was more size dependent (negative slope) when the relative abundance of salmon was greater offshore than nearshore.

Size-dependent vulnerability of salmon and the timing of predation

The relative vulnerabilities of juvenile salmon to predators were size dependent, but the nature of the dependency was a function of both prey and predator sizes. The vulnerability of juvenile salmon to predation by small planktivores declined sharply with increasing salmon length; whereas, vulnerability to large planktivores increased over the same range of salmon lengths (Fig. 6). Vulnerabilities of salmon to both small and large piscivores declined with increasing salmon length. Bootstrap estimates of the standard errors of prey vulnerabilities were all less than 0.007. All vulnerability estimates were significantly different from each other, except vulnerabilities of 7-cm salmon to small and large piscivores. Confidence intervals were too narrow to be shown in Fig. 6.

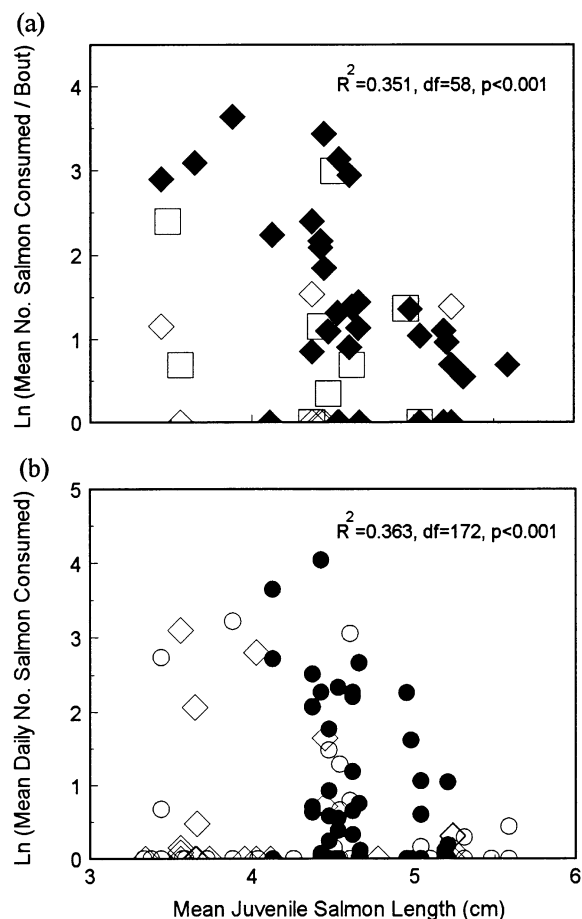
The generally high degree of correspondence between observed and expected length distributions of

Figure 4. Relationship between juvenile salmon distribution, R_a (offshore > nearshore, solid circle; offshore = nearshore, open circle; offshore < nearshore, open diamond), mean juvenile salmon length, and the probability that salmon were present in the stomachs of planktivores, piscivores and invertebrate feeders as predicted from logistic regression analysis. Vertical bars represent the 95% confidence intervals on the estimated probabilities. Note different probability scales used for planktivores and invertebrate feeders.



salmon in predator stomachs (Fig. 7) indicated that this approach yielded a relatively accurate estimate of prey vulnerability functions for salmon exposed to these four predators. For subsequent simulation purposes, non-linear least squares regression models were generated for length-specific vulnerabilities of salmon preyed upon by small planktivores ($Y = 0.964e^{-1.539x}$, $R^2 = 0.934$, d.f. = 4, $P = 0.002$) and large planktivores ($Y = -0.017^{2x} +$

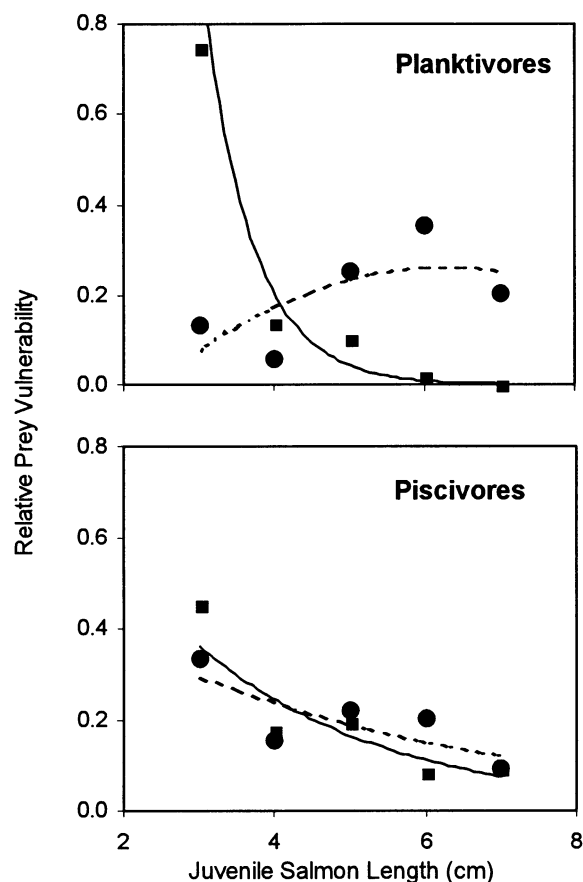
Figure 5. (a) Relationship between mean juvenile salmon length and the mean number of juvenile salmon (natural logarithm transformed) consumed by planktivores (open squares), piscivores (solid diamond) and invertebrate feeders (open diamond) during feeding bouts. (b) Mean daily number of salmon consumed (natural logarithm transformed) by planktivores, piscivores and invertebrate feeders in relation to mean juvenile salmon length and juvenile salmon distribution, R_a (offshore > nearshore, solid circle; offshore = nearshore, open circle; offshore < nearshore, open diamond).



$0.217x - 0.419$, $R^2 = 0.462$, d.f. = 4, $P = 0.140$), small piscivores ($Y = 2.04e^{-0.444x}$, $R^2 = 0.866$, d.f. = 4, $P = 0.010$), and large piscivores ($Y = 0.708e^{-0.247x}$, $R^2 = 0.626$, d.f. = 4, $P = 0.062$).

Simulation modelling of vulnerabilities of salmon to four predator groups during the initial 60 days of sea life indicated that the timing of predation events affected the relative importance of size upon ocean entry and subsequent growth in determining the salmon's vulnerability to predators. However, the size- and growth-dependent vulnerabilities of salmon differed more among predator groups than between May

Figure 6. Length-specific vulnerabilities of juvenile salmon preyed upon by small (<30 cm, solid squares and lines) and large (>30 cm, solid circles and dashed lines) planktivores and piscivores. All vulnerabilities were significantly different from each other except for 7 cm juvenile salmon preyed upon by piscivores.



and June (Fig. 8). Overall, salmon reared to a large size in hatcheries (up to 6 cm) were substantially less vulnerable to predators than wild salmon that entered the ocean at lengths ranging from 2 to 3 cm. When simulated predation was shifted from May to June, the vulnerability of both stocks of salmon to predators became more growth dependent, but the timing of predation events was more important for wild salmon that always entered the ocean at a relatively small size. When predation was applied in May, size upon ocean entry and subsequent growth were equally important in determining the vulnerability of wild salmon to predators, but when predation was applied in June, growth rates of wild salmon were more important than initial size. Juvenile salmon entering the ocean at lengths exceeding 3.5 cm were relatively invulnerable to predation by small planktivores in June. The vulnerability of salmon to large planktivores exhibited a maximum for salmon

Figure 7. Parameters used in analyses of length-specific vulnerabilities of juvenile salmon preyed upon by (a) small and large planktivores, and (b) small and large piscivores. Expected length distributions of salmon in predator stomachs were calculated using prey vulnerabilities derived from an optimization algorithm (see Figure 6).

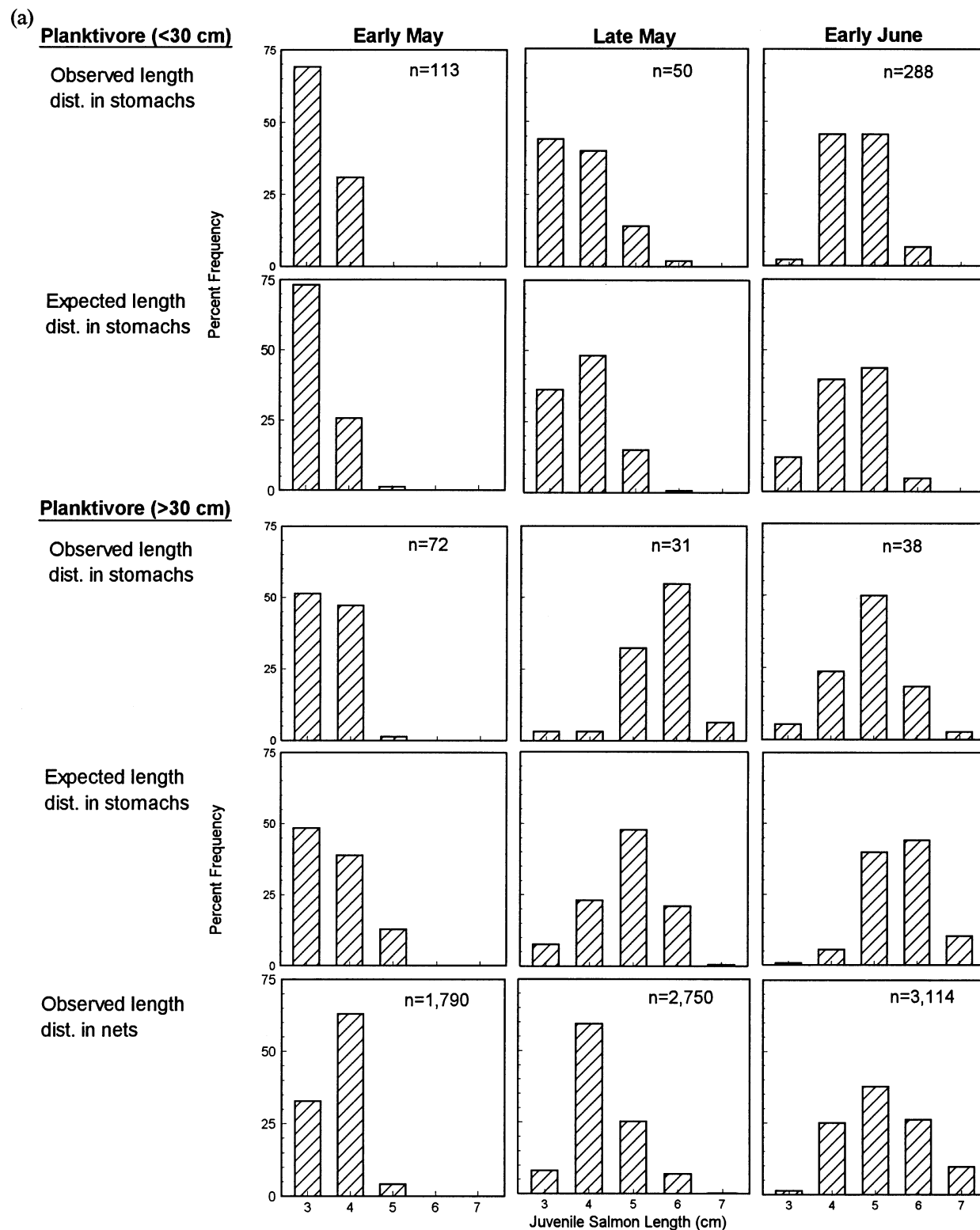
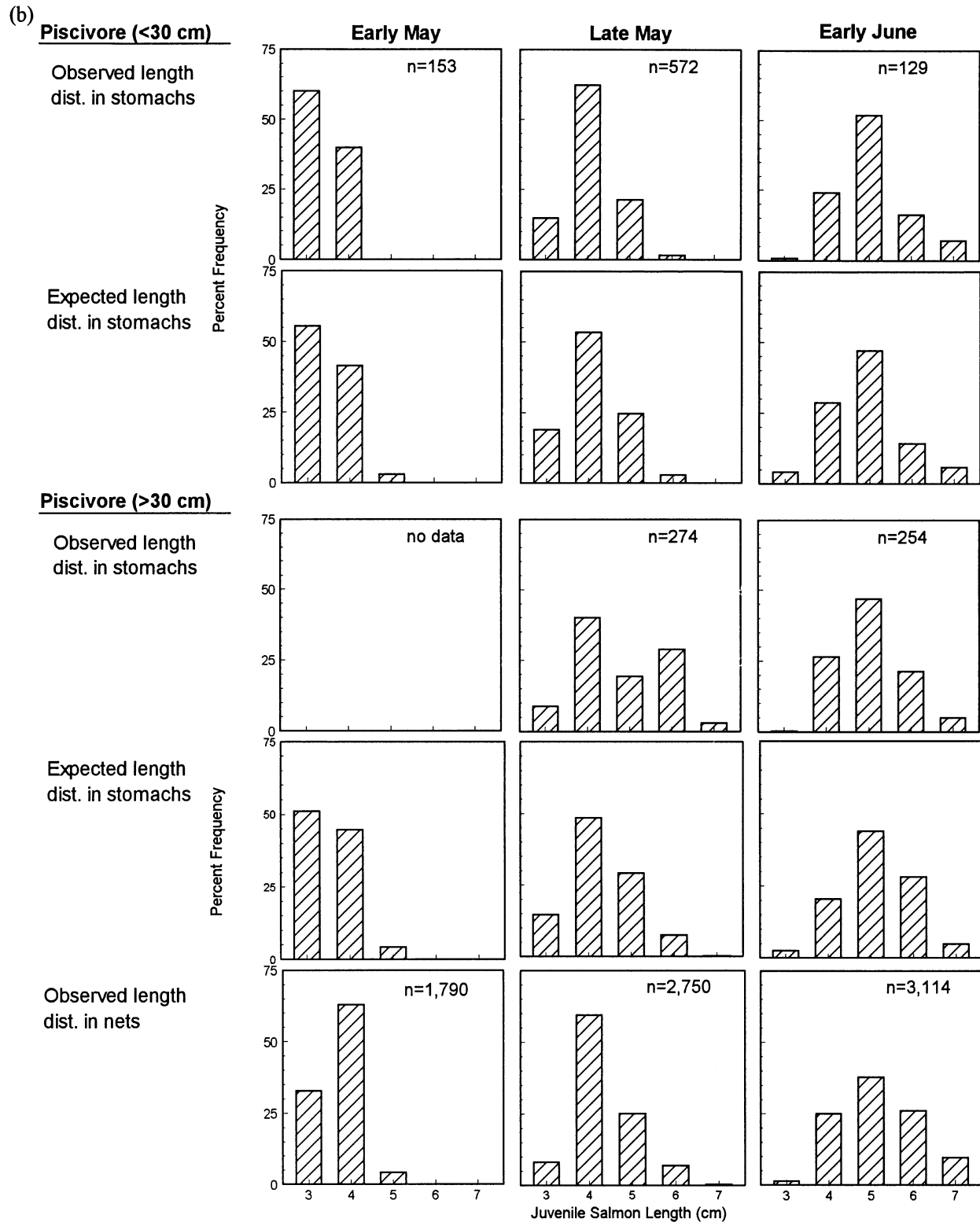


Figure 7. *Continued*

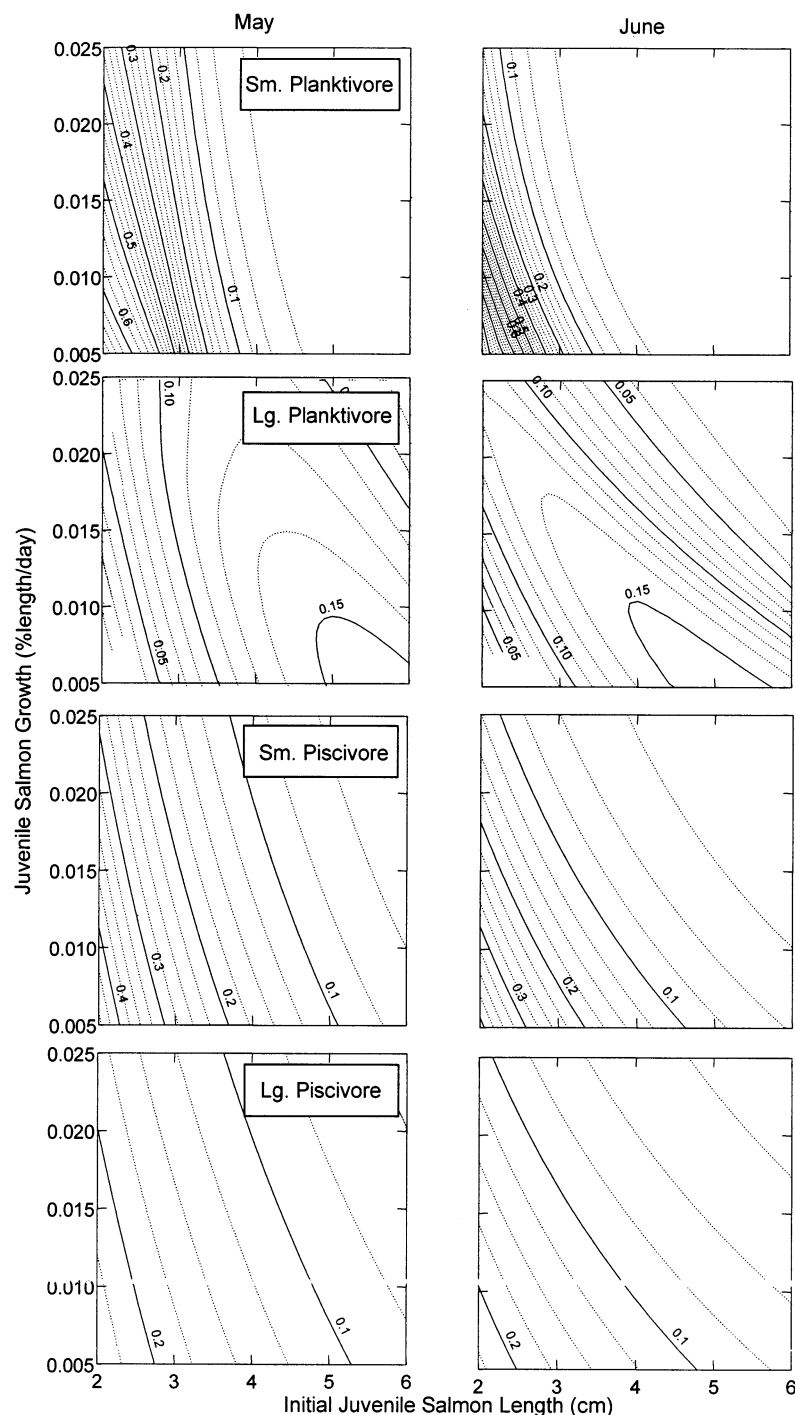


Figure 8. Mean vulnerabilities (over the initial 60 days of sea life, 1 May–30 June) of juvenile salmon to small (<30 cm) and large (>30 cm) planktivores and piscivores in relation to length of salmon upon ocean entry and salmon growth rate when simulated predation is applied in May vs. June.

entering the ocean at 5–6 cm length, and this maximum was shifted to smaller initial sizes and was more strongly growth-dependent when predation occurred in June. Overall, vulnerabilities of salmon to small predators were more dependent on the salmon's initial size and growth than was their vulnerability to large predators.

DISCUSSION

All fish groups examined in PWS fed to some extent on juvenile salmon, but mean daily individual predator consumption of salmon was generally low. Trout and gadids consumed the greatest numbers of juvenile salmon per day on average (1–2 day⁻¹) while other fishes consumed

$< 1 \text{ day}^{-1}$. Estimates of daily individual predator consumption of salmon may be biased high, because data from samples collected between 21:00 and 09:00 h were used. Although, there were no significant differences in consumption of salmon between this 12-h period and the full 24-h day in 1995, mean consumption tended to be higher during the 21:00–09:00 h period. However, this source of error should not affect comparisons of numbers of salmon consumed among predator groups.

The majority of the diets of all fishes were composed of prey other than juvenile salmon. Significant alternative prey included macrozooplankton for herring, pollock (Willette *et al.*, 1999a) and pelagic rockfishes; other fishes for trout and gadids; and benthic invertebrates for demersal rockfishes and shallow benthic fishes (Table 8). The relatively small fraction of salmon in predator diets suggested that predators rarely became satiated from feeding solely on salmon.

Juvenile salmon faced an increasing probability of encountering predators in nearshore habitats from early May to June (Table 6). Seasonal increases in relative abundance of Dolly Varden trout were probably due to emigration from overwintering habitats in numerous lakes bordering PWS (Armstrong and Morrow, 1980). Less pronounced increases in the relative abundances of immature pollock may have been due to initiation of diel vertical migrations associated with warming of the surface mixed layer (Clark and Green, 1990). This increase in predator abundance nearshore may have provided an additional motivation for salmon to disperse offshore in June (Fig. 2).

These results support the hypothesis that low macrozooplankton density leads to dispersion of juvenile salmon from shallow nearshore habitats and greater predation risk, but zooplankton type is an important factor. Juvenile salmon were found to disperse when the biomass of large copepods or the fraction of their diet composed of these prey declined (Fig. 3). This behavioural response was only associated with large copepods indicating that prey type was an important factor affecting feeding rate. Parsons and LeBrasseur (1973) observed that juvenile pink salmon selected large copepods 2–4 mm in length, and juveniles feeding on these prey achieved feeding rates seven times greater than those consuming *Pseudocalanus*. Juvenile chinook salmon also selected prey within this size range, and their growth was positively correlated with the density of this size of prey (English, 1983). My results were consistent with these other studies. In PWS, salmon stomach fullness was significantly lower in shallow nearshore habitats where the fraction of their diet composed of small copepods was greater. Salmon stomach fullness in shallow nearshore

habitats also declined significantly when the fraction of their diet and the biomass of large copepods declined from May to June (Tables 4 and 5). Maintaining high feeding and growth rates likely minimized the salmon's period of vulnerability to predators and increased their probability of reaching the critical size necessary to survive their first winter (Beamish and Mahnken, 1999).

However, dispersion from shallow nearshore habitats in search of food would not increase reproductive success unless either the density or type of food available to juveniles limited growth nearshore. Two independent analyses indicated that growth of juvenile salmon in PWS was limited when juvenile densities were high. Willette *et al.* (1999b) found that growth rates of tagged juvenile pink salmon were reduced when zooplankton density was low and the number of juveniles in hatchery release groups exceeded 20 million. Similarly, Paul and Willette (1997) found that energy content of juvenile salmon was lower in areas near hatcheries, where juvenile salmon densities were high compared with areas distant from hatcheries in eastern PWS. These results suggest that salmon density is an important factor affecting behavioural responses of salmon to changing prey abundance. In the present study, sampling was conducted at various sites intended to provide a range of salmon densities, but the majority of the sites were near hatcheries, where salmon densities were probably higher than elsewhere in PWS. In areas where salmon densities are much lower and there is less competition for food, salmon would be expected to disperse from nearshore habitats at a lower threshold of copepod biomass.

The notion that shallow nearshore habitats provided salmon a refuge from predators was supported by my analysis of predation losses. A roughly five-fold increase in the probability of occurrence of salmon in predator stomachs when salmon dispersed (Fig. 4) was evidence that they were vulnerable to a greater number of predators. Higher predation on salmon when they dispersed was due to greater overlap between predator and prey, not an increase in the number of salmon consumed per feeding bout (Fig. 5a). Salmon that dispersed from shallow habitats at a small size were most vulnerable, because salmon size was inversely related to both the probability of occurrence of salmon in predator stomachs and the number of salmon consumed per feeding bout.

Predation by piscivores and invertebrate feeders likely occurred inshore of the 20-m isobath as salmon dispersed from shallow nearshore habitats. As previously noted, visual shoreline surveys estimated the relative abundance of salmon in waters only a few meters deep, so R_a indicated changes in salmon relative abundance between habitats offshore of the 20-m isobath and very shallow nearshore habitats. The majority of piscivores

and invertebrate feeders captured in this study were associated with kelp beds that typically extended out to the 20-m isobath. Salmon that remained in waters only a few metres deep minimized their exposure to predators, but when they moved out over the kelp beds, they were silhouetted against the sky during daylight hours and were within relatively close proximity to potential predators. Further from shore, water depths in western PWS increased dramatically, essentially separating salmon from demersal fishes.

My analysis indicated that planktivores also consumed greater numbers of salmon when salmon dispersed offshore (Fig. 5b). However, since this behaviour was inversely correlated with the biomass of large copepods, it was not possible from this analysis to determine if the observed correlation was in fact due to dispersion of salmon offshore or prey switching (Willette *et al.*, 1999a). Both processes probably affected predation losses of salmon to planktivores.

My results supported Healey's conclusion (Healey, 1982) that relatively intense size-dependent mortality occurred when juvenile chum salmon migrated out of shallow nearshore habitats, because size-dependent predation losses of salmon increased when salmon dispersed from these habitats (Figs 4 and 5). But, this behaviour was not simply an ontogenetic migration, as described by LeBrasseur and Parker (1964), because it was only weakly related to date period (Fig. 2) and was not related to salmon length (Fig. 3). The strong size-dependent mortality that occurs when juveniles disperse from nearshore habitats is therefore probably not limited to the 4.5–5.5 cm length range as described by Healey (1982).

My analysis of vulnerabilities of salmon to predators supported the notion that predation on salmon was size dependent, but the nature of the size dependence was a function of both predator and prey sizes. The relative size dependence of encounter rate and capture probability determines the shape of the prey vulnerability function (Bailey and Houde, 1989; Juanes, 1994). Peak vulnerabilities at small prey sizes (Fig. 6) indicated that capture success rates for small planktivores were strongly size dependent or encounter rates were weakly size dependent or both. Peak vulnerabilities at intermediate prey sizes indicated that capture success rates for large planktivores were relatively independent of prey size (over the range of salmon lengths used in the analysis) or encounter rates were more strongly size dependent or both. The slightly more size-dependent vulnerability function for small than large piscivores was consistent with stronger size-dependent capture success rates for the smaller predator. These prey vulnerability functions (Fig. 6) were consistent with results from other studies (Miller *et al.*, 1988) that have demonstrated that capture success rates

were more strongly size dependent when prey-predator size ratios were large (i.e. when prey were a large proportion of predator size). In this case, capture success decreases rapidly as prey size approaches a limit set by the mouth gape of the predator. Popova (1978) concluded that the maximum size of fish taken by piscivores was about 50% of the predator's length. Examination of the length distribution of the four predator groups used in my analysis (small planktivores: range 15–23 cm, mean = 22 cm; large planktivores: 30–74 cm, mean = 49 cm; small piscivores: 14–30 cm, mean = 26 cm; large piscivores: 30–47 cm, mean = 37 cm) indicated that the largest salmon taken by the small predators were near the maximum size typically consumed by fishes, so a strong size-dependent capture success function should be expected. Further, the considerable difference between the prey vulnerability functions for the two size classes of planktivores may have been due to a greater difference between the sizes of the small and large planktivores compared to the two size classes of piscivores. The stronger size-dependent vulnerabilities for small planktivores compared with small piscivores may have resulted because planktivores were inexperienced with highly evasive large prey (Werner *et al.*, 1981; Vinyard, 1982) at a time when their diets were dominated by less evasive zooplankton (Willette *et al.*, 1999a).

In my analysis of prey vulnerabilities, data were partitioned into three 2-week periods to estimate a single prey vulnerability function that fit the data across all periods simultaneously. An exponential increase in juvenile salmon body weight during early sea life led to an exponential decline in potential individual predator consumption of salmon due to the limited size of predator stomachs (Petersen and DeAngelis, 1992). Aggregating the data over short time periods minimized this effect, which would have otherwise tended to skew length distributions of salmon observed in stomachs toward smaller sizes. The generally high degree of correspondence between observed and expected length distributions of salmon in predator stomachs (Fig. 7) indicated that this approach yielded a relatively accurate estimate of prey vulnerability functions for salmon exposed to these four predators, because a single function for each predator reproduced the size distribution of salmon in their stomachs over three date periods.

Salmon reared to a large size at hatcheries before their release into the marine ecosystem are presumably less vulnerable to predators during their initial 60 days of sea life than wild salmon, which enter the ocean at smaller sizes (Fig. 8). In an analysis of historical survival of tagged salmon released from PWS hatcheries, Willette *et al.* (1999a) found that size of salmon at release was slightly more important than subsequent juvenile growth

in determining survival to adult. The results from the present analysis generally support this conclusion with the exception that salmon reared to a large size may be more vulnerable to large planktivores (Fig. 8) if these predators are present in the system at the time of release. However, salmon that have been reared to a large size are typically released in June, when abundances of large planktivores (primarily adult pollock) are declining in the surface layer (Table 6).

My results generally supported the hypothesis that the timing of predation modifies the nature of size-dependent predation losses of salmon to different predator groups. When simulated predation was shifted from May to June, the vulnerability of salmon to predators became more growth dependent (Fig. 8). But, the size- and growth-dependent vulnerabilities of salmon differed more among predator groups than between May and June, suggesting that changes in the composition of predator fields could more strongly affect the nature of size-dependent predation losses than changes in the timing of predation losses. The taxonomic and size composition of predator fields can change from year to year (Table 7) due to large-scale changes in predator distribution or variable recruitment success among predator groups. Vulnerabilities of salmon will be strongly dependent on initial salmon size when predator fields are dominated by small planktivores and piscivores, and much less so when predator fields are dominated by larger animals.

The timing of predation was potentially more important for wild salmon, which always entered the ocean at relatively small sizes (Heard, 1991). The timing of predation could be altered by changes in overall predator abundance, availability of alternative prey, or salmon behaviour affecting the overlap between predator and prey. Increasing abundances of immature pollock and trout in nearshore habitats from May to June (Table 6) could increase the importance of earlier growth by shifting predation risk later in the season. Conversely, declining abundances of adult pollock in offshore habitats (Table 6) could decrease the importance of growth by shifting predation risk earlier in the season. Changes in the timing and duration of the bloom of large copepods (Eslinger *et al.*, 2001; this volume p. 81) could have similar effects by influencing availability of alternative prey (Willette *et al.*, 1999a) or salmon distribution (Fig. 3) or growth (Willette *et al.*, 1999b). A lack of persistence from year to year in correlations between survival and salmon size upon ocean entry or juvenile growth (Holtby *et al.*, 1990; Willette, 1996) could thus be explained by changes in the composition of predator fields or the timing of predation losses during the juvenile lifestage. Future research should focus on how interactions among

these processes determine the nature of size-dependent predation losses of wild and hatchery stocks of salmon.

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